8. Resonance

Measure the peak; do not merely hunt for one

Prove the preferred rhythm, then test its electrical counterpart

Lesson 8 · two sessions · learner: pendulum; adult: low-voltage electrical sweep

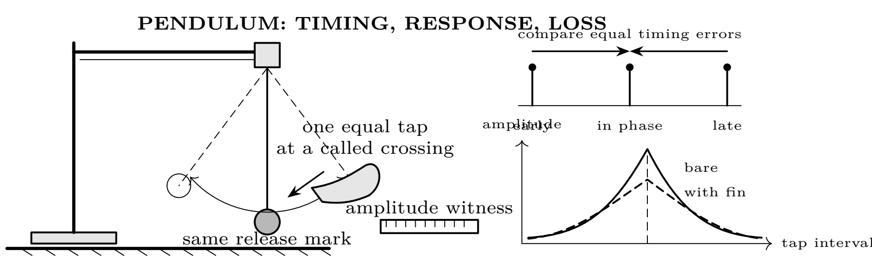
What you need

- Stand, string, small mass, ruler, tape flag
- Stopwatch and partner to call crossings
- Two identical cardstock damping fins
- Adult-controlled, isolated signal generator
- $1\,\mathrm{k\Omega}$ current-limiting resistor
- Known film capacitor and inductor; AC meter or oscilloscope

Predict first

Equal taps are delivered slower than, at, and faster than a pendulum's own rhythm. Sketch amplitude versus tap rate. Where is its maximum? Now add a cardstock fin: predict what changes about the maximum's height and width. Finally predict how doubling C changes an LC circuit's preferred frequency.

A mechanical standard of proof



1. **Measure the free rhythm.** Release from the same small tape mark, without a push. Time 12 complete cycles from a same-direction centre crossing. Repeat four times. For each trial calculate

$$T_i = \frac{t_{12,i}}{12}, \qquad \bar{T} = \frac{T_1 + T_2 + T_3 + T_4}{4}, \qquad f_{\text{mech}} = \frac{1}{\bar{T}}.$$

   The four T_i values must span no more than 5% of $\bar{T}$. If they do, repeat after checking the pivot, release mark, and cycle call.
2. **Calibrate one tap.** A partner calls the same-direction centre crossing. Practise a short sideways finger tap with identical travel each time. Do not chase the mass with the finger. Reset from the tape mark for every run.
3. **Sweep both sides.** Apply eight taps, separated by

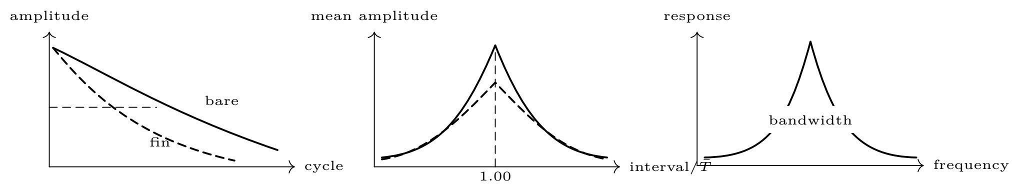
$$0.70\bar{T}, \quad 0.85\bar{T}, \quad 1.00\bar{T}, \quad 1.15\bar{T}, \quad 1.30\bar{T}.$$

   Measure the largest displacement after tap eight. Randomize the order of the five settings; repeat the whole set three times. Random order helps expose practice, fatigue, and a drifting tap size.

interval	$0.70\bar{T}$	$0.85\bar{T}$	$1.00\bar{T}$	$1.15\bar{T}$	$1.30\bar{T}$
amplitude run 1 (cm)					
amplitude run 2 (cm)					
amplitude run 3 (cm)					
mean amplitude (cm)					

Loss is evidence, not an inconvenience

4. **Measure free decay.** With no taps, release from amplitude A_0. Count cycles until the amplitude first falls below $A_0/2$; call that $N_{1/2}$. Repeat three times. Attach one fin and repeat, then two fins.
5. **Sweep with damping.** Repeat the five timing settings with one fin. Use the same release, number of taps, and randomized order. Plot bare and damped mean amplitudes on the same axes.



FREE-DECAY CHECK | TWO-SIDED PEAK | HEIGHT AND WIDTH

Questions before the electrical analogy

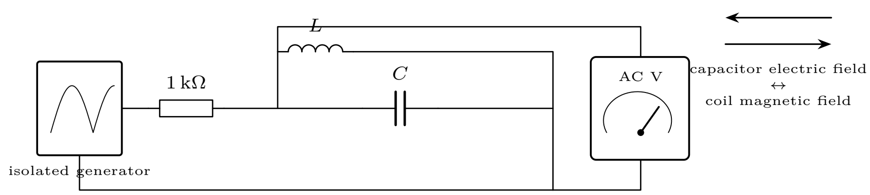
1. Which timing produced the greatest *mean* response? Did response fall on both sides, or did you merely stop searching at the largest point?
2. Did the best interval agree with $\bar{T}$ within one tested step?
3. Were early and late settings equally far from $1.00\bar{T}$ similar? If not, what uncontrolled asymmetry could explain it?
4. Did fins reduce $N_{1/2}$? Did they lower and broaden the driven peak?
5. Which conclusion is supported: "resonance makes energy," or "phase controls how efficiently repeated taps transfer supplied energy"?

Mechanical diagnostics

Observation	Check, then repeat
No preferred interval	Tap cadence, tap travel, pivot friction, and sweep range.
Maximum at an endpoint	Extend the sweep; an endpoint cannot establish a peak.
Runs climb in order	Operator learning or release drift; randomize the next set.
Peak moves after adding fin	Fin changed length or mass position, not only drag.
Large left/right mismatch	Crossing call, ruler parallax, or unequal phase error.

Translate the test to an LC circuit

ADULT-CONTROLLED, CURRENT-LIMITED RESPONSE SWEEP



Predict, calculate, sweep, repeat

1. Record the component values and tolerances. Calculate
$$f_{\text{calc}} = \frac{1}{2\pi\sqrt{LC}}.$$
Calculate a starting sweep from $0.50 f_{\text{calc}}$ to $1.50 f_{\text{calc}}$. Use at least nine points and place more points near the predicted maximum.
2. With power off, the adult builds the pictured parallel LC network behind the $1\,\text{k}\Omega$ limiting resistor and checks the meter range. Keep generator amplitude fixed and low. The learner records; the adult alone operates and changes the energized circuit.
3. Sweep upward through all frequencies, then downward. At each point wait the same settling time and record tank voltage. A real peak must survive sweep direction. If the two readings differ materially, repeat after checking generator amplitude, connection motion, meter bandwidth, and settling time.

f/f_{calc}	.50	.70	.85	.95	1.00	1.05	1.15	1.30	1.50
V upward									
V downward									
mean V									

4. Let f_{meas} be the frequency of the largest mean voltage. Report the fractional disagreement:
$$\text{difference}(\%) = 100\frac{|f_{\text{meas}} - f_{\text{calc}}|}{f_{\text{calc}}}.$$
Do not call the formula wrong until component tolerance, coil resistance, stray capacitance, meter loading, generator output, and frequency resolution have been considered.
5. Add a known series resistor to the inductor branch and repeat the up/down sweep. The expected damping signature is a lower, wider peak. If f_1 and f_2 are the half-power frequencies from an oscilloscope-quality measurement, estimate
$$Q \approx \frac{f_{\text{meas}}}{f_2 - f_1}.$$
With a simple AC multimeter, describe width qualitatively; do not claim a precision Q its bandwidth and sampling cannot support.
6. Change C by a known factor and repeat. Before measuring, predict the ratio
$$\frac{f_{\text{new}}}{f_{\text{old}}} = \sqrt{\frac{C_{\text{old}}}{C_{\text{new}}}}.$$
For example, four times the capacitance predicts half the frequency.

Electrical diagnostics

Pattern	Interpretation to test
No maximum in range	Arithmetic, units, inductor value, or too-narrow sweep.
Up/down curves disagree	Drift, insufficient settling, loose lead, or source variation.
Maximum shifts with meter	Meter loading is part of the circuit. Record the setup.
Added resistance sharpens peak	Connection or scale changed; this contradicts the intended one-variable test.
Measured f_0 consistently low	Total capacitance or inductance exceeds nominal, often through stray effects.

Final proof

ACCEPT THE RESONANCE CLAIM ONLY WHEN ALL SIX AGREE

1. Four free-period trials agree and define a mechanical prediction.
2. The mechanical response has an interior maximum and falls on both sides.
3. Added drag shortens free decay and lowers or broadens the driven response.
4. The electrical up/down sweeps reproduce an interior maximum.
5. f_{meas} is compared numerically with $1/(2\pi\sqrt{LC})$, with component and measurement limits stated.
6. Changing C moves the maximum in the square-root direction and by a defensible amount.

If any item fails, write *not yet demonstrated*, identify the failed control, and specify the next repeat. A tall reading by itself is not proof.

What the analogy earns

The pendulum and LC circuit share three measured signatures: a natural rate, a two-sided driven-response maximum, and loss-dependent decay and peak width. That supports a common oscillator model. It does not mean charge is a hanging mass or that resonance supplies energy. The driver supplies energy; timing governs accumulation; loss drains it.

BOUNDARY

The learner investigation ends with the pendulum. The electrical sweep uses only an isolated, current-limited, low-voltage generator and is assembled deenergized. The adult controls power and connections. No mains, opened appliances, charged power capacitors, ignition coils, sparks, or exposed high voltage belong in this lesson. Adult status does not imply electrical qualification.

Evidence ledger

Quantity	Mechanical oscillator	Electrical oscillator
Predicted natural rate	$\bar{T} =$ ______, $f =$ ______	$L =$ ______, $C =$ ______, $f_{\text{calc}} =$ ______
Measured maximum	interval = ______	$f_{\text{meas}} =$ ______
Repeatability	period span = ______%	largest up/down difference = ______
Added-loss result	$N_{1/2}$: bare ______, fin ______	peak: original ______, resistor ______
Decision	[ ] demonstrated [ ] not yet demonstrated; next controlled repeat: ______